Release Notes ZEISS INSPECT 2023

Table of Contents

ZEISS INSPECT 2023 Service Pack 5 1
ZEISS INSPECT 2023 Service Pack 4 4
ZEISS INSPECT 2023 Service Pack 3 6
ZEISS INSPECT 2023 Service Pack 2 8
ZEISS INSPECT 2023 Service Pack 1 Patch 1 10
ZEISS INSPECT 2023 Service Pack 1 11
ZEISS INSPECT 2023 13
Notes for Upgrading from GOM Software 2022 15
ZEISS INSPECT Third Party Notices 16
System Requirements 17
Version Information ZEISS INSPECT 2023 18
Necessary Drivers for Sensors 20

ZEISS INSPECT 2023 Service Pack 5

Released at: 2025-02-17 (2023.7.0.87)

Notes for compatibility for later ZEISS INSPECT major versions – ATOS Sensors / ARAMIS Sensors / T-Scan Hawk 2 / X-Ray Systems (such as METROTOM 6, METROTOM 1)

ZEISS INSPECT 2023 Service Pack 5 contains a change which ensures that several sensors can be still operated with ZEISS INSPECT 2023 alternately with later ZEISS INSPECT major versions. Therefor updating to ZEISS INSPECT 2023 Service Pack 5 is highly recommended to avoid future problems when ZEISS INSPECT 2023 (from Service Pack 5 upwards) is to be operated alternately with later ZEISS INSPECT major versions higher than ZEISS INPSECT 2025.

This software version contains several bugfixes and improvements to the prior ZEISS INSPECT 2023 versions, e.g.

- EULA: The prior end user license agreement of ZEISS INSPECT 2023 Software is replaced by a new end user license agreement. Therefor users must accept this end user license agreement, when starting ZEISS INSPECT 2023 Service Pack 5, when updated from an older installed ZEISS INSPECT 2023 version.
- ZEISS Quality Suite: The ZEISS INSPECT 2023 Service Pack 5 includes the ZEISS Quality Suite version 6.2.2479.0.
- New CAD Converter: The ZEISS INSPECT 2025 Service Pack 5 contains a new CAD converter supporting following formats "NX 2406", "CATIA 2024", "3DXML 2024", "Inventor 2025", "JT 10.9", "Revit 2024", "Solid Edge 2024", "Parasolid 36.1", "SolidWorks 2024", "Rhino 8", "Navisworks 2025", "Creo 11".
- ZEISS INSPECT Installer: With ZEISS INSPECT 2023 Service Pack 5 .Net 8.0.12 is included.
- Add-on Editor - metainfo.json: When using the "Save" button on an manually edited metainfo.json file in the Add-on editor, a basic check for the structural correctness of the metainfo.json file is done. With this, problems such as could be avoided caused by invalid metainfo.json files.

- Add-on Editor: Now, when some unnamed content (such as unnamed UI-Configurations or Project Keyword sets) no longer an error message is displayed when opening the Add-On Editor. Instead, this unnamed content will be loaded properly. With this a possible crash is fixed, when opening projects containing unamend project keyword sets while the Add-on Editor is open.
- ATOS Sensors with big measuring volumes - Distance Indicator: A problem is fixed when using ATOS Sensors with big measuring volumes (MV1000 and higher) for manually scanning in combination with a visible distance indicator. In prior ZEISS INSPECT versions this could lead to connection lost to the image acquisition server.
- Stage projects / Virtual Clamping - "New Stage": A bug has been fixed that caused a software crash when creating a new stage in an existing virtual clamping project.
- Localization of Add-Ons: A possible problem is fixed, which caused in prior ZEISS INSPECT versions, that not all texts in dialogs were displayed in the correct language even if the texts are translated properly.
- PONTOS Live: Software no longer crashes when executing a reference point measurement using system ARAMIS SRX sensor with ARAMIS Controller.
- Photogrammetry / Scale Bars / Computation: The handling of outside points in a scale bar changed for outlier computation to avoid possible uncomputed scale bars in photogrammetry measurement series. Affected projects need to be triggered once with a change for the "Reference Points" in the "Acquisition Parameters" for photogrammetry.
- VMR projects - Recalculate Project / Scan Coverage: A possible software crash is fixed, when recalculating certain projects when "Do you want to check whether there are enough scan data for your elements?" is set to "Yes" in the "Scanning Template and Project Settings" dialog to enable the check for scan coverage.
- Import project file: A problem is fixed, when importing project files (*.zlements, exported with File > Export > Elements > Selected Elements) into existing projects. In prior ZEISS INSPECT 2023 versions the software could crash when importing certain *.zelements files.
- ARAMIS Controller - Deformation tracking / SCPI Server: Now, the software synchronizes itself to an controller clock (but not to Windows system clock) during deformation tracking, and timestamps can be written therefor correctly to the log file (*.csv).
- Measuring systems - Compatibility with later ZEISS INSPECT major versions: ZEISS INSPECT 2023 Service Pack 5 contains a change which ensures that several sensors can be still operated with ZEISS INSPECT 2023 alternately with later ZEISS INSPECT major versions. Therefor updating to ZEISS INSPECT 2023 Service Pack 5 is highly recommended to avoid future problems when ZEISS INSPECT 2023 (from Service Pack 5 upwards) is to be operated alternately with later ZEISS INSPECT major versions higher than ZEISS INPSECT 2025.
- Correlate: The software no longer crashes when creating new projects based on certain project templates containing nominal surface components.
- "Surface Component From Point Grid" - "Edit Creation Parameters": A possible crash is fixed when using "Edit Creation Parameters" on the surface component from the point grid elements in certain projects.
- ZEISS INSPECT Kiosk / Optical Systems - Gray value features / Mesh adaption: Gray value features that cannot be successfully blended into the mesh, and are required for the computation of Virtual Clamping, RPS alignment or have active checks/inspections, may affect results in virtually clamped or temperature compensated state, see Known Issues below. For this reason, the Kiosk now detects these situations and displays a warning message after evaluation, before showing the report and confirmation dialog. In asynchronous evaluations, Kiosk marks affected projects as "failed," while inline evaluations add identified elements to the warning list and generate a warning signal to the external PLC. A warning message is also included in the project keywords for both evaluation types.

Covered Cases

CNR-221405-H7Y8M4, CNR-178579-T7L3R6, CNR-188588-G9K2T5, CNR-189174-Y4K5K6, CNR-212697-K6C8R1, CNR-188788-W5B0D7, CNR-190114-X8F6X9, 203776, CNR-202397-N0Z0H7, CNR-190940-K1F5P7, CNR-190565-H2P7V9

Known Issues

Virtual Clamping, Temperature Compensation, Gray Value Feature with Mesh Adaptation, Alignment, Checks on Gray Value Feature, Link to Actual with Mesh Adaptation

Gray value features or imported actual features must be blended into the mesh to be accurately considered by mesh operations like Virtual Clamping or RPS alignment in virtually clamped state.

When the measuring principles "Gray Value Feature with Mesh Adaptation" or "Link to Actual with Mesh Adaptation" are applied, the software creates additional elements, such as fitting circles, based on the adapted mesh borders around the features. If mesh adaptation fails, a fitting element is still created as a duplicate of the actual feature, remaining in an unclamped or uncompensated state. Consequently, when these elements are considered in the computation, they can affect virtual clamping or alignment in the virtually clamped / temperature compensated state. Any checks on affected elements will also be based on the unclamped/uncompensated state.

ZEISS INSPECT 2023 Service Pack 4

Released at: 2024-09-25 (2023.6.0.104)

This software version contains several bugfixes and improvements to the prior ZEISS INSPECT 2023 versions, e.g.

- ZEISS Quality Suite: The ZEISS INSPECT 2023 Service Pack 4 includes the ZEISS Quality Suite version 6.1.2462.0. With this also CodeMeter 7.60d will be installed.
- ZEISS INSPECT Installer: With ZEISS INSPECT 2023 Service Pack 4 .Net 8.0.8 is included.
- Add-on - Script Migration from GOM Software 2022: Now, when upgrading from GOM Software 2022 or prior to ZEISS INSPECT 2023 Service Pack 4 during the migration of user-defined scripts (user folder and public folder) into Add-ons the information is stored properly by adding the respective tag "migrated" and "user" or "shared" in the metainfo.json. This applies only if before no older ZEISS INSPECT 2023 version was started on the system. With this, also the migration of ATOS Kiosk Interface settings/customization is easier in case that no older ZEISS INSPECT 2023 version was started on the system before (see Further Information for ZEISS INSPECT 2023 Service Pack 4 below).
- Project Recalculation / Undo - Point clouds: A very specific crash is fixed, if a recalculation is performed for a project with a point cloud and several alignments.
- Curves / Tolerances from tolerance regions - Tolerances band / Reports: A bug has been fixed that caused that at report pages tolerance bands are not visible if the source is set to "from tolerance region" and the legend is not displayed.
- Curves / Tolerances from tolerance regions - Tolerances band: Now, the tolerance band is visible in the section view, if the tolerance source is "from tolerance region" and legend is visible and the property for uncomputed values is set to transparent.
- Surfaces / Triangulate Geometrical Element - Alignments / Undo: Now, actual surfaces created with "Triangulate Geometrical Element" are taken properly into account when changing the alignment and using "Undo" afterward.
- DoubleRobotCell environments - ATOS Kiosk Interface: Now computation works properly again, when only one reference point size is used.
- VMR - Calibration measurement series / Collisions: Recomputation of the calibration measurement series path is prevented when the measuring setup is invalid. In rare cases this could lead to robot collisions in prior ZEISS INSPECT versions.
- Deformation tracking - ARAMIS 24M: A problem is fixed, which caused in prior ZEISS INSPECT 2023 versions, that the deformation tracking frame rate dropped to 0Hz after long-term measurement.
- Airfoil - Profile Chord Line: A problem is fixed, that led to a crash when opening the "Create Profile Chord Line" again after a profile chord line was created, which uses a non global direction.
- Zeiss Inspect X-RAY - "Orthogonal View": A problem is fixed, which caused under circumstances, that the volume was not displayed in the "Orthogonal View". In prior ZEISS INSPECZ 2023 versions this could happen, if a newly created project was saved without switching to the "Orthogonal View".
- ARAMIS SRX / ARAMIS Controller: The calibration with deactivated tracking spots is now possible.
- X-RAY - Prealignment (Volume) / Additional best-fit: Now, when the option "Compute additional best-fit" is used for "Prealignment (Volume)", the deviation of this best-fit is computed properly and displayed in the dialog as well as in the "Properties" tab if the alignment is selected in the explorer.
- Calibration - ATOS LRX / Cross CC50: A bug was fixed, which in rare cases could have led to problems with the calculation of the automatic exposure times.
- X-RAY - Prealignment (Volume) / Stage projects: A possible crash is fixed for stage projects when recalculating project in combination with using "Prealignment (Volume)".
- Optical3D - Scan measurement series: There is no longer any possibility of longer freezes (sometimes endlessly) of the software in connection with larger projects when cut out points or positions are optimized.

- GD&T / Min/Max deviation labels - Stage ranges: A possible crash is fixed for Min/Max deviation labels on GD&T inspections in combination with restricted stage ranges. In prior ZEISS INSPECT versions the software could crash if Min/Max deviation labels on GD&T inspections were used, where the elements, on which the GD&T inspection depends on, are restricted to a specific stage range and the user switches to an stage which is not in that stage range.
- GD&T Flatness - Restricted Portion: Now the software no longer crashes if a flatness is used together with the restriction by diameter. This could happen in rare cases before.
- Defect Analysis - "Distance To Outer Surface": The calculation performance of checks such as "Distance To Outer Surface" has been slightly improved.
- Volume Inspection - Identify Materials: A crash is fixed, which could happen during preview of "Identify Material" command, when using certain volume.
- X-Ray - "Redefine Volume Value Range": A possible software crash is fixed, when aborting the command "Redefine Volume Value Range".
- Add-Ons - Api Settings / File selector: Now, accessing configuration via settings api works now also properly for entries of type "file" (file selector).
- Execute measurement series - Cut out against CAD / Prealignment: A possible problem is fixed, where under specific circumstances the measurement execution failed because neither the alignment or the measurement series can be computed in certain projects. For such existing projects a remeasuring or reimport of the scanned data is needed to correct it.
- "Analyse Profile Form And Position": A bug has been fixed that allowed to calculate a form and position analysis even though nominal and actual section are not on the same level.
- GD&T / Flatness tolerances - Element group: A problem is solved, which could lead the software to crash during recalculation, when a flatness check with SZ on an element group is existing.

Covered Cases

CNR-182341-V1G7W8, CNR-176562-D9Q8H4, CNR-180740-R5W5Y7, CNR-184052-L0R2D7, CNR-185379-H3J5K9, CNR-184602-X2J7N9, CNR-185833-F6V0H3, CNR-182973-W1C4Y8, CNR-186928-Z3K1Y2, CNR-184162-K3R8F1, CNR-186488-S4P0G4, CNR-186040-M2X0L5, CNR-187148-T4F0G1, CNR-187680-R6S9K9, CNR-185542-H1H3K9, CNR-189224-H9P7B6

Known Issue

ATOS Sensors with big measuring volumes: There is a problem when using ATOS Sensors with big measuring volumes (MV1000 and higher) for manually scanning in combination with a visible distance indicator, that could lead to connection lost to the image acquisition server. As workaround disable the distance indicator under “View” > “3D View Elements” > … .

Further Information for ZEISS INSPECT 2023 Service Pack 4

Migrating ATOS KioskInterface settings/customizations from GOM Software 2022 to ZEISS INSPECT 2023

Updating from GOM Software 2022 to ZEISS INSPECT 2023 Service Pack 4 with a prior configured ATOS Kiosk Interface should be done by following steps:

- Start ZEISS INSPECT 2023 Optical 3D
- Copy public folder to 2023 by using “Copy to current version” on the start screen
- Open the Add-on Editor, select the “shared” add-on and press “Finish Editing”
- Run the “Setup” for the KioskInterface - this will create an Add-on named “KioskInterface (Customization)”

ZEISS INSPECT 2023 Service Pack 3

Released at: 2024-07-22 (2023.5.0.188)

This software version contains several bugfixes and improvements to the prior ZEISS INSPECT 2023 versions, e.g.

- ZEISS Quality Suite: The ZEISS INSPECT 2023 Service Pack 3 includes the ZEISS Quality Suite version 5.3.2023.0.
- "Volume Defect Element (from Volume)": Now, the command "Volume Defect Element (from Volume)" needs less memory and calculates faster than in prior ZEISS INSPECT 2023 versions.
- ZEISS INSPECT Installer: With ZEISS INSPECT 2023 Service Pack 3 .Net 8.0.6 is included.
- X-Ray - Close Project: A possible problem is fixed, that lead in prior ZEISS INSPECT 2023 versions under circumstances to application hangs when closing projects and/or existing the application.
- "Start" workspace - "Recently Used Projects" / "Project Templates": Now, the preview images for projects under "Recently Used Projects" / "Project Templates" in the "Start" workspace are loaded faster than in prior ZEISS INSPECT 2023 versions.
- X-Ray - VoluMax9 Titan: Now when starting a movement with the joystick while a prior movement started by using the software is not yet finished, the joystick movement no longer overrides the current software movement command.
- Add-On Editor - "External Folder": A bug in the Add-on Editor is fixed, that prevented to execute gom.api.settings related code when the Add-on was connected as "External Folder".
- VMR / Measuring Setup - Imported Pre-Measured Reference Points: A bug is fixed, which could have led to wrong positions of pre-measured reference points during "Edit Creation Parameters" after executing "Transform Reference Points To Fixture" or "Transform Measuring Object To Fixture", because creation parameters were not transformed correctly.
- Polygonize and Recalculate - Gray value Feature (Mesh Adaption): Now, when mesh adaption is executed implicitly during "Polygonize and Recalculate" for gray value Feature (Mesh Adaption) elements the result is the same as when the mesh adaption is executed by calling the command "Operations" > "Mesh" > "Other" > "Perform Mesh Adaption". In prior ZEISS INSPECT 2023 version in the first case the mesh was not adapted properly to all "Gray value Feature (Mesh Adaption)" holes as it was the case when executing "Perform Mesh Adaption" to create an adapted mesh variant.
- GD&T Datum-System - ASME Y14.5: Now, when creating a datum system based on two symmetric planes for datum 1 facing the same direction, the computation mode ASME Y14.5 computes the datum correctly. This was broken before.
- GD&T Surface Profile: When using a nominal patch compound and actual point cloud together for the creation of a surface profile, the software does not crash anymore.
- Scan measurements - Fine Alignment: A bug is fixed, which could have led to wrong transformation of measurement positions after deleting measurements and recalculating afterwards.
- GD&T Flatness tolerance - GD&T Filter: When changing the cutoff value of the filter for a flatness tolerance fast and erratically with the mouse wheel the software might crash in prior ZEISS INSPECT 2023 versions. This bug is fixed.
- Copy & Paste - Properties / Shape-related dimension: If an element is copied whose value for "Shape-relevant dimension" was previously changed under properties, the properties for the copied element are set correctly now.
- VMR - "Create Equally Distributed Measurement Positions": When using "Create Equally Distributed Measurement Positions" a preliminary mesh is created properly when executing the measuring series. In prior ZEISS INSPECT versions in that case a final mesh (but without a real polygonization) was created by mistake instead of a preliminary mesh.
- Eagle Eye Measurements / Caligo gmxlz files: A bug is fixed where only the eagle eye measurement was imported to ZI while importing a gmxlz file. Now all files are imported.
- X-RAY - "Identify Materials": The computation time of the command "Identify Materials" is improved.

- Add-On Editor - Dialog: Now, when inserting several dialogs as "Separate dialog file" into an Add-On the names of newly inserted dialog files are generated properly. In prior ZEISS INSPECT versions this was not always the case, and the prior inserted dialog file could be overwritten by newer ones.
- Eagle Eye Measurements - Import: A bug is fixed where all imported scans from an eagle eye measurement could have the same name. Now, only unique names are existing because they are numbered.
- "New Project" - "Acquisition Parameters" / "Measuring setup stability check": Now, for projects created with "New Project" the defaults for the "Checks" in the "Acquisition Parameters" are the same again as in prior GOM Software 2022 versions. With this the "Measuring setup stability check" in the "Acquisition Parameters" is enabled for projects created with "New Project".
- Add-Ons - Api Settings: A bug is fixed regarding the metainfo.json of the Settings API, that prevented overlaying entries of being accessed and parsed via script. Additionally, a problem is fixed, which caused an application crash when values smaller than the minimum were set for a parameter in the metainfo.json file.
- Motion Replay: A bug is fixed, which could have led to not executed measurements of a measurement series, if they were recorded at the same position as the previous measurement.
- ISO 3D labels - Save project: Now, when a project is saved the status of ISO 3D labels is saved properly, too. In rare cases it could have happen that a change of the ISO label did not get saved in prior ZEISS INSPECT 2023 versions.
- X-Ray - Create Volume Defects (from Volume) / Find inclusions: Now, when using the defect type "Find inclusions" when executing Create Volume Defects (from Volume) works properly. in prior ZEISS INSPECT 2023 versions under circumstances also defects of type "Find defects" were found by mistake.
- Airfoil - Profile Edge Circles: A bug is fixed that caused the fitting of a profile leading edge circle not worked when using a camber line extrapolated with absolute length. Instead, a touch circle was used.
- VMR - "Check Fixture Position": The script "Check Fixture Position" is improved. For example, a problem is fixed, that the script ends in a possible endless loop, when it recognized at first properly a wrongly positioned fixture and the check was started again with "Retry" after the position of the fixture was corrected.
- Gray Value Feature (Mesh Adaption) / Height offset - Fitting element: In cases where the gray value feature result needs to be projected back into the plane aligned with the outer sheet (e.g. multilayered sheet metals), the fitting element based on the mesh boundary used to be projected into the nominal element. However, this behavior has now been modified in ZEISS INSPECT 2023 Service Pack 3, so that the result will be projected onto the plane the gray value feature is inspected on. One known issue arising from this change is that the fitting element becomes reliant on the internal plane the gray value feature is inspected on. The result in normal direction of this plane will remain unaffected by mesh operations such as virtual clamping.
- ATOS Sensors - Sensor firmware: With ZEISS INSPECT 2023 Service Pack 3 a new sensor firmware for GOM Scan 1.
- ATOS Kiosk Interface - Check Fixture Position: A problem is fixed when using the fixture position check in combination with deleting the fixture position elements (KeepCheckFixturePositionElements = False). In prior ZEISS INSPECT 2023 versions the part contains only a preliminary mesh while the polygonized mesh was moved into the clipboard.
- VMR - Calibration measurement series / Collisions: Editing the calibration measurement series is prevented when the measuring setup is invalid. In rare cases this could lead to robot collisions in prior ZEISS INSPECT versions.
- Scripting: Now, importing script archives (created with GOM Software 2019 and prior) into ZEISS INSPECT 2023 leads to an error instead of a crash. Importing such script archives is no longer supported by ZEISS INSPECT 2023. Instead, Add-ons or packages have to be used.

Covered cases

CNR-131810-R5P8R1, CNR-176614-H5D2J1, CNR-177183-Z3N9X7, CNR-177202-B2P6V3, CNR-177452-P3W4N0, CNR-177936-Z7W8T9, CNR-177939-S0Q3F4, CNR-178263-N9T0S9, CNR-178362-B9K7L7, CNR-178516-R2P5W7, CNR-180522-D5Q2V8, CNR-180523-N1T8X6, CNR-181727-P6D2Z3, CNR-183787-Z1L2X3, CNR-184854-G9C0R6, CNR-182477-D0N0L7

ZEISS INSPECT 2023 Service Pack 2

Released at: 2024-05-06 (2023.4.0.459)

Notes for compatibility for later ZEISS INSPECT major versions – ZEISS ScanCobot

For ZEISS INSPECT 2023 Service Pack 2 the corresponding "ZEISS Hardware Service (Remote)" driver contains a change that allows to downgrade it from a later ZEISS INSPECT major version. Therefor updating to "ZEISS INSPECT 2023 Service Pack 2" is highly recommended to avoid future problems when ZEISS INSPECT 2023 (from Service Pack 2 upwards) is to be operated alternately with later ZEISS INSPECT main versions.

This software version contains several bugfixes and improvements to the prior ZEISS INSPECT 2023 versions, e.g.

- Reporting/Table: The "Show All Elements" setting of the table widget is now restored by reports after saving and loading the project again. This setting is also restored by reports from old projects, or projects created with project templates of prior GOM Software versions. With this, report pages using tables with activated "Show All Elements" display all inspections, when e.g. editing or recalculate them. In prior ZEISS INSPECT 2023 versions in that case the table content was cleared.
- Stage projects - "Import Project" / Mesh adaption: A possible bug is solved, when importing projects with performed mesh adaption as new stage into a project with an performed mesh adaption. In prior ZEISS INSPECT versions this led by mistake to an additional actual mesh, when the names of meshes differs between the imported and the target project.
- ZEISS INSPECT Installer: With ZEISS INSPECT 2023 Service Pack 2 .Net 6.0.29 is included.
- Polygonize And Recalculate Project - Gray value Feature (mesh adaption): A problem is fixed, where "Polygonize And Recalculate Project" did not compute mesh adaption.
- X-Ray acquisition: When user wants to set voltage higher than allowed Max. high voltage without startup, a warning is displayed in the control panel.
- X-Ray acquisition: A fix was made to avoid resetting of the reconstruction cuboid when the scanning trajectory/path is changed.
- X-Ray - Volume reconstruction: Vertical-field-of view extension and reconstruction strategy as 'Reduced volume(user-guided)' or 'Reduced volume(automatic)' combination was allowed and resulted in failed scan. After the fix user cannot start scan with this combination anymore.
- "Polygonize And Recalculate Project": "Polygonize And Recalculate Project With Report Pages" now proper recalculates all elements, alignments and report pages when Gray value Feature (Mesh Adaption).
- GD&T / Orientation plane: A bug is fixed while creating a position check for a point with a circular tolerance zone, now the "Orientation plane" is correctly pre-filled and and true positions are set correct now.
- Export Table Content - Expressions: When exporting a table with expressions containing the "<" or ">" character, the table is now exported correctly. Before, in that case values computed by an expression containing "<" or ">" character, were missing in the exported file.
- ZEISS INSPECT 2023 Installer: Now, when using the parameter /INSTALL_UPDATE_SERVICE (YES or NO) for the ZEISS INSPECT 2023 installer via command line, this is forwarded to the included ZEISS Quality Suite Installer, which is executed in case that no or an older ZEISS Quality Suite than included to the used ZEISS INSPECT installer is installed on the system. With this, disabling the ZEISS Quality Suite Update Service is also possible by using only the ZEISS INSPECT installer on a clean system.
- Add-on Editor: Scripts: Now, when closing the add-on editor while running an script, wich was started inside the editor, the software does not crash. In prior ZEISS INSPECT 2023 versions this could happen under circumstances.
- License management - Network licenses / Viewer: Now, starting the "Viewer" no longer blocks network licenses from an used license server.

- ARAMIS Controller: A new firmware is provided with ZEISS INSPECT 2023 Service Pack 2. The controller must be restarted after executing the firmware update.
- Identify Adapters - Vector adapter definitions with offset: Now, the command "Identify Adapters" uses the defined offset of vector adapters.
- "Intersection Curve From Geometrical Element": A possible problem is fixed, which caused that for curves too few curve points were created, when created with with "Intersection Curve From Geometrical Element" using cones. This happened especially, when the cone used for intersection has a very small angle. In prior software versions this led to unexpected or wrong results when such curves are used for inspections. For existing curves created in prior software versions an "Edit Creation Parameter" may be necessary to get the correct results.
- ATOS Sensors - Sensor firmware: With ZEISS INSPECT 2023 Service Pack 2 a new optional sensor firmware for several ATOS sensors (ATOS 5, ATOS 5X, ATOS LRX, ATOS Capsule) is provided. With this a new camera firmware is included.
- GD&T / Envelope Principle - Alignment: A bug was fixed which led to alignment dependent results for envelope conditions when a symmetry plane is involved in the Datum System. If the flatness of one of two planes is significantly worse than the flatness of the other plane and the dimension between the planes is greater than the dimensions of the planes the symmetry plane have had a problem. Affected projects need to be recalculated once.
- ATOS Kiosk Interface - Multipart evaluation: A part naming problem in the evaluation workflow has been fixed which could result in broken projects with an invalid mesh hierarchy.
- Multi-part measurement projects: A possible crash is fixed, when creating measuring setups for multiple parts e.g. with "Automatically Create Elements For Scanning Multiple Parts".
- ZEISS ScanCobot: For ZEISS INSPECT 2023 Service Pack 2 the corresponding "ZEISS Hardware Service (Remote)" driver contains a change that allows to downgrade it from a later ZEISS INSPECT major version. Therefor updating to "ZEISS INSPECT 2023 Service Pack 2" is highly recommended to avoid future problems when ZEISS INSPECT 2023 (from Service Pack 2 upwards) is to be operated alternately with later ZEISS INSPECT main versions.
- Add-ons - Label templates: When a label template of an add-on which is valid for two or more element types is edited for only one element type and the new created template is added to the add-on, the new template includes all element types of the originally template and not only the edited label of one element type.
- Add-ons: Now, add-ons with an additional binary content and a '.' at the end of the name can be loaded.
- Motion Replay: A bug is fixed, which could have led to a non-functioning Motion-Replay measurement, if the controller name was changed.
- ATOS KioskInterface - "Async" mode: Additional tolerances now proper respected by the KIOSK evaluation. Before some projects could be falsely marked as failed.
- Non-part-based projects - Deleting measurements: A bug is fixed which led to a Software crash when deleting the last scan measurements from an ATOS measurement series.
- GD&T checks - Short-wave pass filters: The visual position of the tolerance zone now roughly coincides with the position of the fitting plane. The bug causing an incorrect position of the tolerance zone for planes with short-pass filters has been fixed.
- ATOS KioskInterface: Creating a kiosk user and sharing kiosk configuration between windows users via scripting now proper works again.
- Check Dimension (Dimensional tolerancing): The bug causing incorrect interpretation of the dimensional and geometric tolerancing flag with ISO 20457, leading to wrong tolerance display, has been fixed. The tolerance flag is now correctly interpreted, displaying the expected tolerance value.
- Table - Edit table template: Now, when editing a existing table template choosing a new name is not required anymore as in prior GOM Software 2022.
- GD&T position check for planes: Now, when creating an position check for a plane the filters are enabled in the position tolerance dialog.

Covered cases

CNR-103995-C7V3Z1, CNR-106037-W7Z5V2, CNR-148237-M5K9S0, CNR-166876-Y3Y2Z1, CNR-167260-K7W3G3, CNR-168106-K1F9D2, CNR-168125-H0V0K7, CNR-169175-X1Z9V8, CNR-169446-T0P8P8, CNR-169726-V6C6Y4, CNR-171703-N8M9Z7, CNR-172148-B1V5Q2, CNR-172461-B2N0G6, CNR-172535-P4K7B2, CNR-172561-J5N6T3, CNR-173052-R6C1B8, CNR-173366-R9Q1T4, CNR-173785-Q9Z3F9, CNR-174044-C9C7J5, CNR-174197-S6B1T6, CNR-174408-J3X1Z1, CNR-174708-N9V3S8, CNR-174735-G6D1Q3, CNR-175126-G5R3W4, CNR-175228-K5X2H0, CNR-175430-Y3C2P4, CNR-175455-Y3C8H8, CNR-175655-M8V4B7, CNR-175733-W5D1X2, CNR-175801-L1H6V4, CNR-175877-F5P5B2, CNR-176107-L6D6B3, CNR-176118-K1Y7W2, CNR-176150-V9Z9D7, CNR-176288-X5F0J9, CNR-176459-T7X3J4, CNR-176517-K8M5W5, CNR-176532-B6Z4F3, CNR-176580-F9G2N9, CNR-176624-T1J0X9, CNR-177103-F9H5X2, CNR-177195-M9N0W2, CNR-177497-N9Y7F5, CNR-177549-D3T6L8, CNR-177669-M0V5Z1, CNR-177786-M5D2Y6, CNR-177857-Z4Q1D0, CNR-178124-H3J8R2, CNR-178133-K1B3R1, CNR-178518-V8D9M1, CNR-178788-W1K2M6, CNR-179619-X1L3M4, CNR-179647-W3M7Q6, CNR-180389-Q7J7L7, CNR-180665-K8C8Y2, CNR-83533-H7G3M0, CNR-174880-X3P0L9

ZEISS INSPECT 2023 Service Pack 1 Patch 1

Released at: 2024-02-26 (2023.3.1.3)

With ZEISS INSPECT 2023 Service Pack 1 Patch 1 a problem with possible wrongly transformed measurements is fixed, when measuring with VMR in particular when not using photogrammetry. This version is therefore highly recommended for all VMR users.

Covered Case

CNR-160703-J6N0C1, CNR-173534-N8M5W7, CNR-174157-D7K1X9, CNR-174179-M8S3F3, CNR-173909-W2H9T3, CNR-174744-H3J1D3, CNR-174974-H8X5L1, CNR-176035-P7Z6M3

ZEISS INSPECT 2023 Service Pack 1

Released at: 2024-02-09 (2023.3.0.969)

This software version contains several bugfixes and improvements to the prior ZEISS INSPECT 2023, e.g.

- ZEISS Quality Suite: The ZEISS INSPECT 2023 Service Pack 1 includes the ZEISS Quality Suite version 5.2.1990.0.
- New version of Image Acquisition driver (14.00.008)
- New .Net 6.0.26
- ATOS Sensors: A new sensor firmware for several ATOS Sensors (ATOS-5, ATOS-5X, ATOS LRX, ATOS Capsule).
- ARAMIS Controller: ARAMIS Controller: A new firmware is provided with ZEISS INSPECT 2023 Service Pack 1.
- VMR - "Measure with Rotation Table": A bug is fixed, which could have led to a real hardware collision when inserting measurements using the rotation table dialog with the sensor in a collision position.
- Project consistency check: With ZEISS INSPECT 2023 Service Pack 1 a consistency check is activated for the creation of projects from a template and when importing data from another project. It can happen that projects with inconsistent data cannot be loaded anymore. In case this happens please contact the support.
- ATOS KioskInterface: A problem is fixed, when updating from a prior in GOM Software 2022 configured ATOS KioskInterface to ZEISS INSPECT 2023. In the prior ZEISS INSPECT 2023 version, after copying the "Public folder", running the "Setup" and transferring the customization from the migrated Add-on "shared" to the Add-on "KioskInterface (Customization)" (see Known Issues below) it was not possible to start the Kiosk without deleting the "KioskInterface" folder and its content from the "shared" Add-on. Now, deleting the "KioskInterface" from the "shared" Add-on is no longer necessary in this case.
- Start Screen - Sample Data: Now, when clicking the first time on the sample data tile of start screen the user can choose the folder, where the sample data (downloaded from the software store) was stored. This folder may be changed later within the "Preferences" in the "User folder" section.
- Camber line - I-Inspect: A bug is fixed when creating a camber line using I-Inspect with "Measuring principle" "Transformed Camber line" for the actual camber line. In prior ZEISS INSPECT 2023 versions, the creation of the actual camber line caused an error. Furthermore, only the nominal element was created and the local coordinate system and the form and position element were removed.
- Inspection / Check "Volume" - Patch compounds: A bug was fixed, which could lead under certain circumstances to an incorrect calculated volume check on a patch compound.
- Add-on Editor: Opening the Add-on editor no longer stalls the software when some unnamed content exists. Instead, an error message is displayed. As example if an "UI Configuration" with an empty name exists the error message is "Cannot open the Add-on Editor because the category (file:///perspectives/) has content for which the name is missing. Assign a name to this content". So, in this case open "View" > "Screen Setup" > "UI Configuration" > "Manage UI Configuration", search for the respective configuration and assign a name to it.
- Sensor firmware for ATOS Core, ATOS Compact Scan, ATOS Triple Scan: ZEISS INSPECT 2023 Service Pack 1 includes the sensor firmware for the older ATOS Sensors such as ATOS Core / ATOS Compact Scan/ ATOS Triple Scan again. With this updating sensor firmware for these sensors is possible, it this should be necessary.
- And several other changes for bug fixes and stability enhancements…

Covered cases

CNR-101707-R8S8C7, CNR-119800-Y1D0R7, CNR-143683-K4V6Y3, CNR-146269-D8Y7F3, CNR-147478-K7D2T4, CNR-153427-T1P4C1, CNR-153493-B0N0D6, CNR-160664-L8C9H3, CNR-167690-C3Y5M1, CNR-168197-M2S7F7, CNR-168463-D3N6H0, CNR-168499-H5Z5H3, CNR-168594-B5J1Y5, CNR-168835-K4R9Y6, CNR-168994-V2Z4T1, CNR-169038-B3M5N8, CNR-169218-X7H4F5, CNR-169385-B0X6N5, CNR-169640-R6J0Z1, CNR-171444-G1X7T8, CNR-171771-Q8G5Z2, CNR-173464-F4S7T7, CNR-93595-F6J3M5, CNR-145139-F9Q2V7, CNR-166890-F1Y7G7, CNR-169418-K5L0B2, CNR-169504-X5X1Y9, CNR-169660-H5H3B2, CNR-170014-T9L4Z8, CNR-170055-M0V7F4, CNR-170257-T2Y3W1, CNR-171717-L8D1H1, CNR-172482-R3B7X3, CNR-173377-F4C9Q5, CNR-46127-D5F9R0

Known Issues for ZEISS INSPECT 2023 Service Pack 1

Updating of ZEISS INSPECT via ZEISS Quality Suite

- When updating the ZEISS INSPECT via ZEISS Quality Suite directly after updating the ZEISS Quality Suite itself to 5.2.1990.0, the update may end under circumstances with the error "Another ZEISS INSPECT installer is already running" (error code 3). In this case, please restart the system and update ZEISS INSPECT again.

METROTOM 1 / METROTOM 6 scout

- When the capture volume dialog is open, moving the zoom axis across the threshold 190 mm will cause the cuboid to jump vertically. So, defining the cuboid on one side of 190 mm and trying to use it on the other side of 190 mm will fail.
- When using the position measuring object command in the area > 190 mm with a small object yielding a measurement position < 190 mm, the automatically determined reconstruction cuboid in the capture volume dialog will be unusable.

In case this happens please contact the support.

Migrating ATOS KioskInterface settings/customizations from GOM Software 2022 to ZEISS INSPECT 2023

Updating from GOM Software 2022 to ZEISS INSPECT 2023 with a prior configured ATOS Kiosk Interface should be done by following steps:

- Start ZEISS INSPECT 2023 Optical 3D
- Copy public folder to 2023 by using "Copy to current version" on the start screen
- Run the "Setup" for the KioskInterface - this will create an Add-on named "KioskInterface (Customization)"
- Open the Add-on editor and transfer the old customization from the "shared" Add-on into the "KioskInterface (Customization)"
    - Select the "KioskInterface (Customization)" in the Add-on explorer and enter the editing mode with "Edit"
    - Delete from the scripts folder the "KioskInterface" folder
    - Select the scripts > KioskInterface folder of the "shared" Add-on in the Add-on explorer
    - Cut this KioskInterface folder and paste it into the "scripts" folder of the "KioskInterface (Customization)" add-on.
    - Leave the editing mode of the "KioskInterface (Customization)" add-on with "Finish editing"

ZEISS INSPECT 2023

Released at: 2023-11-13 (2023.2.0.1520)

ZEISS INSPECT forms the core of the ZEISS Quality Suite and is available in four industry-specific variants. These are pre-configured, ready to use for the application and challenge at hand, and new features have been added with the 2023 release:

Overarching **basic functions of ZEISS INSPECT**

- Increased compatibility for import and export
- GD&T enhancements
- Improved cluster function
- New CAD Converter supporting following formats NX 2212, Parasolid 35, Solid Edge 2023, Solid Works 2023, Catia V5-6 R2023 Parasolid 35.1, Inventor 2024
- New apps available
    - PiWeb reporting plus for template-based reporting even for big samples
    - Autosurfacing to convert scan data into an accurate CAD model

ZEISS INSPECT Optical 3D for inspection and analysis of 3D measurement data

- Software-based movement compensation speeds up data acquisition by tracking of reference points. This means fewer repetitions during measurements.
- New algorithms for gray value features
- GPU acceleration for ATOS Q and ATOS LRX

ZEISS INSPECT X-Ray for analysis of volume data

- Multiview: flexible layout for volume visualization and more
- Region of Interest supports the efficient segmentation of parts into e.g. individual components or different regions, that can be analyzed with different tolerances
- Advanced defect detection and evaluation

ZEISS INSPECT Correlate for the analysis of motion sequences and deformations

- Extensive evaluation functionalities: 6DoF analyses, trajectories, virtual extensometers, combination of scan data with motion measurement and much more.
- Contour detection: for identification of specific deflection points in space and time, e.g. for expanding holes or airbag deployment tests

ZEISS INSPECT Airfoil with specific features for blade geometries

- 3D Throat Area: allows efficient analysis of the minimum 3D area between two blades, whether it is a partial area or a closed channel.
- Virtual Balancing: to calculate the blade center of gravity and derive the unbalance of all individual blades. This function is used to determine the optimal sequence of blades on the disk to generate the lowest possible unbalance.

Want to learn more about this year's ZEISS INSPECT software? Click here (ZEISS ID required) for an overview.

Notes for upgrading from GOM Software 2022

Please refer additionally the section Notes for upgrading from GOM Software 2022, because the ZEISS INSPECT 2023 contains compared with GOM Software 2022 several new firmware and drivers.

Known Issues for ZEISS INSPECT 2023

- Measuring with T-SCAN hawk 2: Changing the scan mode may cause crashes under specific circumstances. To prevent this the image acquisition driver of version 14.00.007 can be installed. This driver is available in the driver archive for ZEISS INSPECT 2023, that can be downloaded here (ZEISS ID required).

Notes for Upgrading from GOM Software 2022

Licenses

The ZEISS INSPECT 2023 requires a new license. To update your license from GOM Software 2022 to the new ZEISS INSPECT 2023, you can use this link License Update. For a simple update of your license to 2023 please leave the comment field empty to allow automatic processing of the dongle update. For further information refer the ZEISS Quality Tech Guide (ZEISS ID required).

Firmware and Drivers

ZEISS INSPECT 2023 contains compared with GOM software 2022 several new firmware and drivers

- New version of CodeMeter coming with the new ZEISS Quality Suite v5
- New version of the GigE image acquisition driver
- New version of ARAMIS 24M Emergent driver
- New version of colin3D driver
- New sensor firmware / camera firmware for several ATOS Sensors (ATOS Capsule, ATOS 5, ATOS 5X, ATOS Q, ATOS LRX, ATOS Airfoil, GOM Scan 1)

Known behavior when updating drivers

- When updating in the ZEISS Quality Suite the GigE image acquisition driver to version 14.00.006 then an error named 1638 may appear. Reason: the installer for the image acquisition driver tries to install a Microsoft Visual Studio C++ Redistributable and if a newer one is already installed the error 1638 is return. This error can be ignored.
- When updating in the ZEISS Quality Suite the ARAMIS 24 driver from 2.36.02 to 2.44.02, the first “update” will uninstall the previous version and only a following “Install” installs the correct driver version.

Additional notes for scripting

The new Add-on Editor replace the Script editor.

For the enhancements of ZEISS INSPECT 2023 including new software functionalities the software is reworked in many areas. This may cause slightly different but better results in ZEISS INSPECT 2023 compared to GOM Software 2022, when using a script created with GOM Software 2022 with the same input data. Additionally, within these changes several script commands / parameters are renamed in ZEISS INSPECT 2023. Therefore, it is recommended to use the command “Tools” > “Update Script” in the right mouse button menu inside the script view of the Add-on editor.

Renamed commands / parameters from GOM Software 2022

- the parameter center_edge_draft was renamed to centered_edge_position (used in commands primitive.create_touch_point_edge, primitive.create_touch_point_edge_on_selection, curve.create_curve_from_touch_points_edge, inspection.measure_by_touch_point_edge_direction)
- the parameter invert_calculated_direction_draft was renamed to invert_offset_direction (used in commands section.create_radial_multisection, section.create_radial_multisection)

Information for Inspection Kernel

With the version ZEISS Inspect 2023, we have optimized our calculation methods with regard to bestfit calculation. This affects inspections plans created from version 2023 onwards. Inspection plans, created with previous versions will be computed using the original calculation methods.

The new numerical methods are operating the same class of the smallest deviations as described by PTB.

Release Notes GOM Software 2022

Release Notes for GOM Software 2022 can be found here.

ZEISS INSPECT Third Party Notices

An overview of third-party software used in the product, is documented in ZEISS_INSPECT_2023_FOSS_Disclaimer.html file, which can be found in the installation folder of the ZEISS INSPECT 2023 (C:\Program Files\Zeiss\INSPECT\2023\, when the default installation folder is used). You can also obtain this overview by calling up the “Licensing Information” link in the “About ZEISS INSPECT” dialog and opening “Open Source Licensing Conditions”.

System Requirements

User rights**:** Windows standard user rights are required to use ZEISS INSPECT – Note: Administrator rights are needed for the initial installation.

Required disc space for installation**:** ~ 100 GB

ZEISS INSPECT is designed to process and evaluate large amounts of data. To smoothly create the measurement, documentation, and evaluation of the data as well as the calculations based on it, certain requirements for the underlying system and its hardware components must be considered.

To use the complete ZEISS Quality Suite functionality an internet connection and ZEISS ID is required.

Below you will find both the minimum requirements and the parameters we recommend for an optimal setup:

	ZEISS INSPECT Optical 3D	**ZEISS INSPECT Correlate**	**ZEISS INSPECT Airfoil**	**ZEISS INSPECT X-RAY**
Minimum Requirements				
Processor	Intel Core i3	Intel Core i3	Intel Core i3	Intel Core i3
RAM	4 GB	4 GB	4 GB	8 GB
Graphics boards	OpenGL compatible	OpenGL compatible	OpenGL compatible	OpenGL compatible with at least 4 GB VRAM
Operating system	Windows 10 (64 bit as minimum version 1607) with current security updates			
Recommended Requirements [1]				
Processor	Intel Core i9/Xeon (16 Core)	Intel Core i9/Xeon (16 Core)	Intel Core i9/Xeon (16 Core)	Intel Core i9/Xeon (16 Core)
RAM	16 GB or more	16 GB or more	32 GB or more	32 GB or more
Graphics board [2]	Certified graphics card (CUDA compatible)	Certified graphics card (CUDA compatible)	Certified graphics board (CUDA compatible)	Certified graphics board (CUDA compatible) with at least 6 GB VRAM
Operating system	Windows 10 (64-bit 21H1) with current security updates Windows 11 (64-bit 22H2) with current security updates			

[1] Depends on used applications and projects.
[2] Here in the ZEISS Quality Tech guide (ZEISS ID required) you can find a list of certificated graphic boards.

Version Information ZEISS INSPECT 2023

The table below lists the versions of additional software / drivers which comes with the ZEISS INSPECT Software or is needed by it (depending on licenses, see section Necessary Drivers for Sensors). This software may be installed by using the ZEISS Quality Suite or may be downloaded from the ZEISS Portal for manually installation.

During an installation of ZEISS INSPECT 2023 also ZEISS Quality Suite and Microsoft ASP.Net Core Runtime will be installed if necessary.

ZEISS INSPECT	ZEISS Quality Suite [1]	CodeMeter [2]	ZEISS Hardware Service [3]	Image Acquisition Driver [3]	Camera driver ARAMIS 24M [3]	colin3D [3]	T-SCAN hawk [3]	ZEISS Hardware Service (Remote) [3]	Microsoft ASP.Net Core Runtime [4]
2023 Service Pack 5 2023.7.0.87 Build 2025-02-13	6.2.2479	7.60d	1.0.0.3	14.00.008	2.44.02	9.1.0.36	3.0.5	2023.7.0.87	8.0.12
2023 Service Pack 4 2023.6.0.104 Build 2024-09-18	6.1.2462.0	7.60b	1.0.0.3	14.00.008	2.44.02	9.1.0.36	3.0.5	2023.6.0.104	8.0.8
2023 Service Pack 3 2023.5.0.188 Build 2024-07-08	5.3.2023.0	7.60b	1.0.0.3	14.00.008	2.44.02	9.1.0.36	3.0.5	2023.5.0.188	8.0.6
2023 Service Pack 2 2023.4.0.459 Build 2024-05-06	5.2.1990.0	7.60b	1.0.0.3	14.00.008	2.44.02	9.1.0.36	3.0.5	2023.4.0.459	6.0.29
2023 Service Pack 1 Patch 1 2023.3.1.3 Build 2024-02-21	5.2.1990.0	7.60b	1.0.0.3	14.00.008	2.44.02	9.1.0.36	3.0.5	2023.3.1.3	6.0.26
2023 Service Pack 1 2023.3.0.969 Build 2024-02-01	5.2.1990.0	7.60b	1.0.0.3	14.00.008	2.44.02	9.1.0.36	3.0.5	2023.3.0.969	6.0.26
2023 Release 2023.2.0.1520 Build 2023-10-25	5.0.1945.0	7.60b	1.0.0.3	14.00.006	2.44.02	9.1.0.36	3.0.5	2023.2.0.1520	6.0.23

[1] Bundled within the ZEISS INSPECT 2023 Installer.

[2] Bundled within the ZEISS Quality Suite Installer, which itself is bundled ZEISS INSPECT 2023 Installer – Update/Installation may be done with the installation of ZEISS Quality Suite.
[3] Can be installed via ZEISS Quality Suite when the system is connected to the Internet or when a local update source is used. Otherwise, they may be installed directly using the installers within the corresponding driver archive, which is available here: Software (ZEISS ID required). Company administrators may contact the software support for getting further information about installation using company internal update sources or may refer e.g. the Techguide (ZEISS ID required).
[4] Bundled within the ZEISS INSPECT 2023 Installer – Update/Installation is done directly over the update/installation of ZEISS INSPECT 2023 itself.

Necessary Drivers for Sensors

The following tables contains information about drivers which must be installed for the usage with different sensors (depending on the licenses). Here two cases must be distinguished. The general case (Table 1) and the case when the IPC Driver license is used, which is necessary for ZEISS ScanCobot systems, formerly GOM ScanCobot (Table 2). In case of an existing IPC Driver license, the "ZEISS Hardware Service (Remote)" must be installed instead of the ZEISS Hardware Service and vice a verse.

For the driver versions used to be with specific ZEISS INSPECT 2023 version please refer section Version Information ZEISS INSPECT 2023.

Table 1: General case (IPC Driver license does not exist)

✓ must be installed ⊖ must not be installed The "ZEISS Hardware Service (Remote)" must not be installed	ZEISS Hardware Service	Image Acquisition Driver [1]	Camera driver ARAMIS 24M [2]	colin3D	T-SCAN hawk
• **ATOS Sensors** such as ATOS Q, GOM SCAN 1, ATOS 5, …. (Licenses [3]) • **ARAMIS Sensors** such as ARAMIS SRX, ARAMIS 3D Camera, … (Licenses [4]) • **T-SCAN hawk 2** (License: Sensor Driver T-SCAN hawk 2) • **PONTOS Live** (License Sensor Driver PONTOS Live) • **X-Ray Systems** such as METROTOM 6, METROTOM 1, … (Licenses [5])	✓	✓			⊖
• **ARAMIS 24M** (License: Sensor Driver ARAMIS 24M)	✓	✓	✓		⊖
• **Photogrammetry** (License: Sensor Driver Photogrammetry)	✓				
• **T-SCAN hawk** (License: Sensor Driver T-SCAN hawk)	✓	⊖			✓
• **T-SCAN** (License: Sensor Driver T-SCAN)	✓			✓	

Table 2: IPC Driver license exists

✓ must be installed ⊖ must not be installed The "ZEISS Hardware Service" must not be installed	ZEISS Hardware Service (Remote)	Image Acquisition Driver [1]	Camera driver ARAMIS 24M [2]	colin3D	T-SCAN hawk
• **ATOS Q** • **ATOS Core**	✓				⊖

[1] In the driver archive which may be downloaded from Software (ZEISS ID Required) this driver is named "InstallGigE-VERSION.exe" and located in the "InstallGigE" subfolders.
[2] In the driver archive which may be downloaded from Software (ZEISS ID Required) this driver is named "ARAMIS_24M_Emergent_driver-VERSION.exe" and located in the "ARAMIS_24M_Emergent_driver" subfolders.

[3] Licenses for several ATOS sensors: Sensor Driver ATOS, Sensor Driver ATOS Core 2M, Sensor Driver ATOS Q for eMotors, Sensor Driver ATOS Q, Sensor Driver GOM Scan 1, Sensor Driver ATOS LRX
[4] Licenses for several ARAMIS sensors: Sensor Driver ARAMIS, Sensor Driver ARAMIS 2D
[5] Licenses for several X-Ray systems: Sensor Driver METROTOM 6, Sensor Driver METROTOM 1, Sensor Driver VoluMax 9 titan